BASE CHANGE ALONG AN IDEAL ON IND-ETALE ALGEBRAS HAS A FULLY FAITHFUL RIGHT ADJOINT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file proetale.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the directedness of the category of factorizations through etale algebras, the identification of the colimit modulo the ideal with the given algebra via flatness and a localization argument, the computation of the Hom sets establishing the adjunction, and the closing full faithfulness.

Editorial note. The label lemma-lift-ind-etale names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source proetale.tex –label lemma-lift-ind-etale –start 1437 –end 1516 –local-id 3577 returned RESULT=OK (L3577 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-proetale/candidates-halfA.json (new exclusive pool assigned by the Lead on 2026-10-13).

Editorial note. The proof contains the parenthetical '(We ignore some set theoretical issues in this proof.)'; this is a standard set-theoretic caveat about the size of the category of factorizations and does not omit any step of the mathematical argument, so the statement is counted with the parenthetical recorded here verbatim.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Characterization of directed categories. Categories, the definition of a directed category. Supplies the three properties (nonempty, upper bounds for pairs, coequalizers) checked for the category of factorizations.
- Stability of etale algebras under tensor products, quotients by coequalizers and localization. Algebra, the lemmas that etale algebras are stable under tensor product, that a map between etale algebras is etale, and the lifting of etale algebras along a surjection of rings, together with the lemma on pure ring maps used to identify a localization. Supply the objects of the category of factorizations, the surjectivity of C -> C-bar, the identification of C/IC with C-bar and the lifting of the etale pieces.

Date: stacks-project:proetale.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

- Permanence of the ind-etale property. Same chapter, the lemma that the ind-etale property is preserved under the operations used in the proof. Supplies the flatness of C/IC -> C-bar, which together with the pureness lemma gives the localization statement.
- Full faithfulness of a functor with an adjoint whose counit is the identity. Categories, the lemma that a functor with a left adjoint such that the unit is an isomorphism is fully faithful. Supplies the conclusion that the right adjoint v is fully faithful because the composite of the base change functor with v is the identity.

Source reference. Original lemma 00U2: sources/stacks-chow/tags.txt, source lines 1096–1096. Prerequisite pointer invoked at proof line(s) 1467; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04D1: sources/stacks-chow/tags.txt, source lines 5711–5711. Prerequisite pointer invoked at proof line(s) 1481; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 00U7: sources/stacks-chow/tags.txt, source lines 1101–1101. Prerequisite pointer invoked at proof line(s) 1474; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04PS: sources/stacks-chow/tags.txt, source lines 6125–6125. Prerequisite pointer invoked at proof line(s) 1487; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 002V: sources/stacks-chow/tags.txt, source lines 112–112. Prerequisite pointer invoked at proof line(s) 1463; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 07RB: sources/stacks-chow/tags.txt, source lines 9861–9861. Prerequisite pointer invoked at proof line(s) 1515; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 097M: sources/stacks-proetale/pinned-original-source.tex, source lines 1417–1417. Prerequisite pointer invoked at proof line(s) 1484; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let A be a ring and let $I \subset A$ be an ideal. The base change functor*

$$\textit{ind-étale } A\textit{-algebras} \longrightarrow \textit{ind-étale } A/I\textit{-algebras}, \quad C \longmapsto C/IC$$

has a fully faithful right adjoint v. In particular, given an ind-étale A/I-algebra $\overline{C}$ there exists an ind-étale A-algebra $C = v(\overline{C})$ such that $\overline{C} = C/IC$.

Proof. Let $\overline{C}$ be an ind-étale A/I-algebra. Consider the category $\mathcal{C}$ of factorizations $A \to B \to \overline{C}$ where $A \to B$ is étale. (We ignore some set theoretical issues in this proof.) We will show that this category is directed and that $C = \operatorname{colim}_{\mathcal{C}} B$ is an ind-étale A-algebra such that $\overline{C} = C/IC$.

We first prove that $\mathcal{C}$ is directed (Categories, Definition 002V). The category is nonempty as $A \to A \to \overline{C}$ is an object. Suppose that $A \to B \to \overline{C}$ and $A \to B' \to \overline{C}$ are two objects of $\mathcal{C}$. Then $A \to B \otimes_A B' \to \overline{C}$ is another (use Algebra, Lemma 00U2). Suppose that $f, g : B \to B'$ are two maps between objects $A \to B \to \overline{C}$ and $A \to B' \to \overline{C}$ of $\mathcal{C}$. Then a coequalizer is $A \to B' \otimes_{f,B,g} B' \to \overline{C}$. This is an object of $\mathcal{C}$ by Algebra, Lemmas 00U2 and 00U7. Thus the category $\mathcal{C}$ is directed.

Write $\overline{C} = \operatorname{colim} \overline{B_i}$ as a filtered colimit with $\overline{B_i}$ étale over A/I. For every i there exists $A \to B_i$ étale with $\overline{B_i} = B_i/IB_i$, see Algebra, Lemma 04D1. Thus $C \to \overline{C}$ is surjective. Since $C/IC \to \overline{C}$ is ind-étale (Lemma 097M) we see that it is flat. Hence $\overline{C}$ is a localization of C/IC at some multiplicative subset $S \subset C/IC$ (Algebra, Lemma 04PS). Take an $f \in C$ mapping to an element of $S \subset C/IC$. Choose $A \to B \to \overline{C}$ in $\mathcal{C}$ and $g \in B$ mapping to f in the colimit. Then we see that $A \to B_g \to \overline{C}$ is an object of $\mathcal{C}$ as well. Thus f is an invertible element of C. It follows that $C/IC = \overline{C}$.

Next, we claim that for an ind-étale algebra D over A we have

$$\operatorname{Mor}_A(D, C) = \operatorname{Mor}_{A/I}(D/ID, \overline{C})$$

Namely, let $D/ID \to \overline{C}$ be an A/I-algebra map. Write $D = \operatorname{colim}_{i \in I} D_i$ as a colimit over a directed set I with D_i étale over A. By choice of $\mathcal{C}$ we obtain a transformation $I \to \mathcal{C}$ and hence a map $D \to C$ compatible with maps to $\overline{C}$. Whence the claim.

It follows that the functor v defined by the rule

$$\overline{C} \longmapsto v(\overline{C}) = \operatorname{colim}_{A \to B \to \overline{C}} B$$

is a right adjoint to the base change functor u as required by the lemma. The functor v is fully faithful because $u \circ v = \mathrm{id}$ by construction, see Categories, Lemma 07RB. □